```c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5

sem_t forks[N];
sem_t room;

void *philosopher(void *arg)
{
    int id = *(int *)arg;

    while (1)
    {
        // Thinking
        printf("Philosopher %d is thinking...\n", id);
        sleep(1);

        // Allow maximum N-1 philosophers to compete for forks
        sem_wait(&room);

        // Pick up left fork
        sem_wait(&forks[id]);
        printf("Philosopher %d picked up left fork %d\n",
               id, id);

        // Pick up right fork
        sem_wait(&forks[(id + 1) % N]);
        printf("Philosopher %d picked up right fork %d\n",
               id, (id + 1) % N);

        // Eating
        printf("Philosopher %d is eating...\n", id);
        sleep(2);

        // Put down right fork
        sem_post(&forks[(id + 1) % N]);
        printf("Philosopher %d put down right fork %d\n",
               id, (id + 1) % N);

        // Put down left fork
        sem_post(&forks[id]);
        printf("Philosopher %d put down left fork %d\n",
               id, id);

        // Leave room
        sem_post(&room);
```

```
        printf("Philosopher %d finished eating.\n\n", id);
    }

    return NULL;
}

int main()
{
    pthread_t philosophers[N];
    int id[N];

    // Initialize fork semaphores
    for (int i = 0; i < N; i++)
    {
        sem_init(&forks[i], 0, 1);
    }

    // Allow only 4 philosophers at a time
    sem_init(&room, 0, N - 1);

    // Create philosopher threads
    for (int i = 0; i < N; i++)
    {
        id[i] = i;
        pthread_create(&philosophers[i], NULL,
                       philosopher, &id[i]);
    }

    // Wait for philosopher threads
    for (int i = 0; i < N; i++)
    {
        pthread_join(philosophers[i], NULL);
    }

    // Destroy semaphores
    for (int i = 0; i < N; i++)
    {
        sem_destroy(&forks[i]);
    }

    sem_destroy(&room);

    return 0;
}
```

OUTPUT:

```
(base) avcoe@avcoe-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ gcc Ass4.c -o Ass4
(base) avcoe@avcoe-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$ ./Ass4
```

Philosopher 1 is thinking...
Philosopher 0 is thinking...
Philosopher 4 is thinking...
Philosopher 2 is thinking...
Philosopher 3 is thinking...
Philosopher 1 picked up left fork 1
Philosopher 1 picked up right fork 2
Philosopher 1 is eating...
Philosopher 4 picked up left fork 4
Philosopher 0 picked up left fork 0
Philosopher 3 picked up left fork 3
Philosopher 1 put down right fork 2
Philosopher 1 put down left fork 1
Philosopher 1 finished eating.

Philosopher 1 is thinking...
Philosopher 0 picked up right fork 1
Philosopher 0 is eating...
Philosopher 2 picked up left fork 2
Philosopher 0 put down right fork 1
Philosopher 0 put down left fork 0
Philosopher 0 finished eating.

Philosopher 0 is thinking...
Philosopher 4 picked up right fork 0
Philosopher 4 is eating...
Philosopher 1 picked up left fork 1
Philosopher 4 put down right fork 0
Philosopher 4 put down left fork 4
Philosopher 4 finished eating.

Philosopher 4 is thinking...
Philosopher 3 picked up right fork 4
Philosopher 3 is eating...
Philosopher 0 picked up left fork 0
Philosopher 3 put down right fork 4
Philosopher 3 put down left fork 3
Philosopher 3 finished eating.

Philosopher 3 is thinking...
Philosopher 2 picked up right fork 3
Philosopher 2 is eating...
Philosopher 4 picked up left fork 4
Philosopher 2 put down right fork 3
Philosopher 2 put down left fork 2
Philosopher 2 finished eating.

Philosopher 2 is thinking...
Philosopher 3 picked up left fork 3
Philosopher 1 picked up right fork 2

Philosopher 1 is eating...
Philosopher 1 put down right fork 2
Philosopher 1 put down left fork 1
Philosopher 1 finished eating.

Philosopher 1 is thinking...
Philosopher 0 picked up right fork 1
Philosopher 0 is eating...
Philosopher 2 picked up left fork 2
Philosopher 0 put down right fork 1
Philosopher 0 put down left fork 0
Philosopher 0 finished eating.

Philosopher 0 is thinking...
Philosopher 4 picked up right fork 0
Philosopher 4 is eating...
Philosopher 1 picked up left fork 1
Philosopher 4 put down right fork 0
Philosopher 4 put down left fork 4
Philosopher 4 finished eating.

Philosopher 4 is thinking...
Philosopher 0 picked up left fork 0
Philosopher 3 picked up right fork 4
Philosopher 3 is eating...
Philosopher 3 put down right fork 4
Philosopher 3 put down left fork 3
Philosopher 3 finished eating.

Philosopher 3 is thinking...
Philosopher 2 picked up right fork 3
Philosopher 2 is eating...
Philosopher 4 picked up left fork 4
Philosopher 2 put down right fork 3
Philosopher 2 put down left fork 2
Philosopher 2 finished eating.

Philosopher 2 is thinking...
Philosopher 3 picked up left fork 3
Philosopher 1 picked up right fork 2
Philosopher 1 is eating...
Philosopher 1 put down right fork 2
Philosopher 1 put down left fork 1
Philosopher 1 finished eating.

Philosopher 0 picked up right fork 1
Philosopher 0 is eating...
Philosopher 2 picked up left fork 2
Philosopher 0 put down right fork 1
Philosopher 0 put down left fork 0

```
Philosopher 0 finished eating.

Philosopher 4 picked up right fork 0
Philosopher 4 is eating...
Philosopher 4 put down right fork 0
Philosopher 4 put down left fork 4
Philosopher 4 finished eating.

Philosopher 3 picked up right fork 4
Philosopher 3 is eating...
Philosopher 3 put down right fork 4
Philosopher 3 put down left fork 3
Philosopher 3 finished eating.

Philosopher 2 picked up right fork 3
Philosopher 2 is eating...
Philosopher 2 put down right fork 3
Philosopher 2 put down left fork 2
Philosopher 2 finished eating.

(base) avcoe@avcoe-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~$
```